CityG M L 3

0 B a s e M o d e l

«G M L » T ypes

C o r e (c o r e

x s d)

V e g e t a t io n (v e g e t a t io n

x s d)

L a n d U se (la n d U se

x s d)

Farm G M L ADE v1

3

fa r m -ty p e s

fa r m -h o o k s

fa r m -e n te r p r is e

fa r m -o p e r a tio n s

fa r m -p r o d u c tio n

fa r m -c o r e

fa r m -s u s ta in a b ility fa r m -r isk

fa r m -te m p o r a l

gm l:A bstractG M L gm l:M easureT ype gm l:C odeT ype gm l:A reaT ype

gm l:R eferenceT ype gm l:SurfaceP ropertyT ype gm l:G eom etryP ropertyT ype

gm l:Tim eP eriodP ropertyType

core:A bstractF eature

core:A bstractF eatureW ithL ifespanT ype

core:A bstractTopLevelC ityO bject

co re:A b stra ctS p a ce

core:A bstractV ersion core:A bstractD ynam izer

veg:P lantC over veg:P lantC overT ype core:A bstractG enericA ttribute core:A bstractG enericA ttributeT ype core:A D E O fA bstractSpaceType core:A bstractD ynam izerT ype core:A bstractV ersionT ype

luse:L andU seT ype luse:L andU se

farm:InputU sedT ype

+ inputT ype: gm l_C odeT ype
+ p ro d u ctN am e: strin g [0..1]
+ quantityA pplied: gm l_M easureT ype [0..1]
+ applicationR ate: gm l_M easureT ype [0..1]

farm:Farm B uildingPropertiesT ype

+ buildingT ype: gm l_C odeT ype [0..1]
+ p u rp o se: strin g [0..1]
+ constructionY ear: gY ear [0..1]
+ asso ciated E n terp riseR ef: g m l_ R eferen ceT y p e [0..1]

farm:P ersonInfoT ype

+ firstN am e: strin g [0..1]
+ su rn a m e: strin g [0..1]
+ co n tactN u m b er: strin g [0..1]
+ sex: gm l_C odeT ype [0..1]
+ a g e: in te g e r [0..1]
+ educationL evel: gm l_C odeT ype [0..1]

farm:H ouseholdInfoT ype

+ h o u seh o ld ID: strin g [0..1]
+ n u m b erO fM em b ers: p o sitiv eIn teg er
+ m em ber: farm _H ouseholdM em berT ype [0..*]

farm:H ouseholdM em berT ype

+ m em b erID: strin g [0..1]
+ relationshipT oH ead: gm l_C odeT ype [0..1]
+ w orkedO nH olding: boolean [0..1]
+ offF arm Incom eS ources: string [0..*]

farm:L abourInputT ype

+ totalF am ilyL abourT im e: duration [0..1]
+ totalH iredL abourT im e: duration [0..1]
+ h ired C o n tracto rU sed: b o o lean [0..1]

farm:Farm ProfileT ype

+ leg alS tatu s: farm _ L eg alS tatu sV alu e
+ m arketIntegration: farm _M arketIntegrationV alue

farm:F arm S izeT ype

+physicalSizeA A U: gm l_A reaType [0..1]
+ econom icS ize: farm _E conom icS izeT ype [0..1]

farm:E conom icSizeT ype

+ totalA griculturalO utput: gm l_M easureT ype
+ cropO utput: gm l_M easureT ype [0..1]
+ livestockO utput: gm l_M easureT ype [0..1]
+ calculationR eferenceP eriod: gm l_T im eP eriodP ropertyT ype [0..1]

farm:D iversificationT ype

+ hasO therE conom icA ctivity: boolean
+otherE conom icA ctivityT ype: gm l_C odeT ype [0..*]
+significanceO fO therA ctivity: farm _SignificanceV alue [0..1]
+ shareO fN onA griculturalO utput: double [0..1]

farm:S torageInfoT ype

+ h asS to rag eA b ility: b o o lean
+ storageT ype: gm l_C odeT ype [0..*]
+ isO n F arm S to rag e: b o o lean [0..1]
+ isM odernF acility: boolean [0..1]

farm:T raceabilityInfoT ype

+ lo tID: strin g
+ b lo ck ch ain H ash: strin g [0..1]
+ originP ropertyR ef: gm l_R eferenceT ype [0..1]
+ originA reaR ef: gm l_R eferenceT ype [0..1]

farm:L egalS tatusV alue farm:M arketIntegrationV alue farm:C om m oditySpecializationV alue

farm:SignificanceV alue farm:P roducerT ypeV alue

farm:_G enericApplicationPropertyO fAbstractRuralEnterprise farm:_G enericApplicationPropertyO fH arvestedLot farm:_G enericApplicationPropertyO fRuralProperty farm:_G enericApplicationPropertyO fAgriculturalO peration farm:_G enericApplicationPropertyO fAbstractC ropArea farm:A bstractF arm G enericA ttributeT ype farm:_AD EO fLandU seType

farm:_A bstractR uralE nterprise

+ occupiedA rea: gm l_A reaT ype [0..1]
+ s ta rtD a te: d a te [0..1]

farm:L ivestockE nterprise

+ livestockT ype: gm l_C odeT ype
+ livestockC ategory: gm l_C odeT ype [0..1]
+ n u m b erO fH ead s: p o sitiv eIn teg er
+ feedingP ractice: gm l_C odeT ype [0..1]
+ h asP ro d u ctio n C o n tract: b o o lean [0..1]
+ h asM ark etin g C o n tract: b o o lean [0..1]

farm:A q u acu ltu reE n terp rise

+ sp eciesR aised: strin g [1..*]

farm:F o restry E n terp rise

+ productT ype: gm l_C odeT ype [1..*]

farm:R uralE nterpriseP ropertyT ype

farm:A griculturalO peration

+ operationT ype: gm l_C odeT ype
+ areaA ffected: gm l_A reaT ype [0..1]
+ targetC ropA reaR ef: gm l_R eferenceT ype [0..1]
+ targ etP ro p erty R ef: g m l_ R eferen ceT y p e [0..1]

farm:A griculturalO perationP ropertyT ype

farm:_A bstractC ropA rea

+ c ro p N a m e: strin g
+ cropC ode: gm l_C odeT ype [0..1]
+ p la n tin g D a te: d a te [0..1]
+ producerT ype: farm _P roducerT ypeV alue [0..1]
+ isIrrig ated: b o o lean [0..1]
+ lod0G eom etry: gm l_S urfaceP ropertyT ype [0..1]
+ lod1G eom etry: gm l_S urfaceP ropertyT ype [0..1]

farm:A nnualC ropA rea

+ cro p R o tatio n In fo: strin g [0..1]

farm:P erennialC ropA rea

+ la stP ru n in g D a te: d a te [0..1]

farm:H arvestedL ot

+ q u an tity H arv ested: g m l_ M easu reT y p e
+ h a rv e s tD a te: d a te
+ q u ality G rad e: strin g [0..1]
+ quantityS old: gm l_M easureT ype [0..1]
+ q u an tity S to red: g m l_ M easu reT y p e [0..1]
+ g eo m etry: g m l_ G eo m etry P ro p erty T y p e [0..1]

farm:H arvestedL otP ropertyT ype

farm:R uralP roperty

+ n a m e: strin g [0..1]
+ h o ld erID: strin g [0..1]
+ c a rC o d e: strin g [0..1]
+ ib g eC o d e: strin g [0..1]
+ officialID: string [0..1]
+ landT enureT ype: gm l_C odeT ype [0..*]
+ k e e p sR e c o rd s: b o o le a n [0..1]
+ m ainA griculturalFocus: farm _C om m oditySpecializationV alue [0..1]
+ m ainM arketA ccessT im e: duration [0..1]
+ lod0G eom etry: gm l_S urfaceP ropertyT ype [0..1]
+ lod1G eom etry: gm l_S urfaceP ropertyT ype [0..1]

farm:InsuranceR iskA ssessm entT ype

+ z a rc S c o re: in te g e r [0..1]
+ calcu latio n D ate: d ateT im e [0..1]
+ riskC lassificationK ey: string [0..1]
+ m ethodologyC ode: gm l_C odeT ype [0..1]

farm:Z arcA naliseSoloT ype

+ c o lle c tio n D a te: d a te
+ la y e rD e p th: strin g
+ sa n d C o n te n t: d o u b le
+ clay C o n ten t: d o u b le

farm:Z arcM onitoram entoS ateliteT ype

+ d a te: d a te
+ sa te llite: strin g
+ ndviV alue: float [0..1]

farm:Z arcH istoricoC ultivoT ype

+ p la n tin g D a te: d a te
+ h a rv e s tD a te: d a te
+ c ro p: strin g
+ ilp U sed: b o o lean

farm:insuranceR isk farm:soilA nalysis farm:satelliteM onitoring farm:cultivationH istory

farm:S ustainabilityM etricsT ype

farm:su stain ab ility

farm:_A bstractF arm Tim eSlice

farm:C ropA reaTim eSlice

+ cropG row thS tage: gm l_C odeT ype [0..1]
+ currentN D V I: float [0..1]

+ m e m b e r

1

0..*

+ eco n o m icS ize

1

0.. 1

1

0.. 1

h o o k

1
0..*

1

0.. 1

+ in p u tU se d 1

0.. 1

1 h o o k

0..*

1

0.. 1

+ p ro d u cesL o t

1

0..*

1 h o o k

0..*

h o o k

1

0..*

+ m a n a g e rIn fo

1

0.. 1

+ farm P ro file 1

0.. 1

+ fa rm S iz e

1

0.. 1

+ h o u seh o ld In fo 1

0..*

+ lab o u rIn p u t 1

0.. 1

+ d iv ersificatio n

1

0.. 1

+ e n te rp ris e 1

0..*

1 h o o k

0..*

h a s ty p e h a s ty p e h a s ty p e h a s ty p e
h a s ty p e